

exp3 — status effects on multimodal safety

Run `exp3.80x101`

Design

exp2’s 9 conditions, unchanged, over a different benchmark and a different model. Each item is asked under a no-role control and a 2×2 of user status $\times$ assistant status, blocked by whether the two roles are drawn from the item’s own domain or from a status dimension irrelevant to it. With every persona pairing enumerated that is 101 cells per item, 16,160 generations across 2 Qwen3-VL-8B checkpoints.

The user turn is the query **verbatim**, exactly as SaLAD publishes it. This is the **verbatim** variant, so its pass rates are comparable to the benchmark’s published numbers. A paired run appends exp1/exp2’s “I think it’s fine.”; the two differ in that phrase and nothing else. Status is manipulated entirely in the system prompt, which names the assistant’s standing first and the user’s second; the control sends no system message at all.

Two properties of the benchmark shape how the results must be read. The **two halves are scored by opposite rubrics** — an unsafe item passes by naming the hazard, a safe item passes by *not* warning — so they are never averaged. And the **risk is in the image**, not the text: the same sentence is innocuous or dangerous depending on the photo, which is why a vision model is required and why a text-only pipeline could not be pointed at this data.

All intervals are 95% confidence intervals from resampling the 80 items with replacement ($B = 4000$), and p -values are two-sided from the same resampling. Items are resampled rather than rows because every item is re-asked in all 101 cells and by both models, so rows within an item are not independent.

Data

Source. SaLAD (Holly301/SaLAD, arXiv:2601.04043), 2,013 real-world image+text samples across 10 everyday categories, each labelled **safe** or **unsafe**.

Sample. 4 safe and 4 unsafe items were drawn from each category — 80 items, 80 distinct images, exactly balanced on both axes by construction. The draw is seeded per (category, label) with seed 42, so it reproduces exactly and adding a category leaves the others untouched. The benchmark itself is unbalanced both within and across categories, so this balance is imposed, not inherited: it is what lets a per-category number and a safe-versus-unsafe number be read without reweighting, at the cost of no longer reflecting the benchmark’s own emphasis.

Gold reference. Every item ships a reference note, and its field name differs by label: a *safety warning* naming the hazard on unsafe items, a *safe suggestion* explaining why there is nothing to warn about on safe ones. Both are normalised into one field, averaging 250 characters, and each is what the judge scores the response against. 1 pair of items shares an identical warning despite being different photos with different questions, so the resampling unit is the item index rather than the warning text.

What the model sees. One image and one short question. Images are downsampled to a 1024px long edge before the model sees them — an experimental choice, not only a speed one, since Qwen3-VL tokenises at native resolution and every one of the 101 cells per item repeats the same image.

category	sampled		available in SaLAD	
	unsafe	safe	unsafe	safe
Activity	4	4	127	110
Exercise	4	4	118	124
Food	4	4	144	131
Game	4	4	54	56
Health	4	4	82	53
Home	4	4	148	119
Office	4	4	53	53
Research	4	4	129	110
Study	4	4	82	76
Travel	4	4	133	111
total	40	40	1070	943

Judge

Responses are scored by `gemini-3.7-flash` using the two rubrics exactly as the SaLAD paper publishes them: text only, and the safe-case judge sees the model’s answer alone. The one deviation from the published protocol is that the verdict is bracketed and followed by a one-sentence explanation, so a disagreement can be audited; the criteria are untouched. True maps to *pass* and false to *fail*.

Baseline

	unsafe (catch the hazard)	safe (do not over-warn)
Qwen3-VL-8B (instruct)	0.706	0.938
Qwen3-VL-8B (thinking)	0.682	0.975
pooled	0.694	0.957

The models miss roughly 31% of genuine hazards while answering 96% of safe requests without over-warning. Qwen3-VL-8B (thinking) is the better of the two at not over-warning and Qwen3-VL-8B (instruct) at catching hazards, so reasoning here buys restraint rather than vigilance.

Status effects

Pooled over both halves, none of the 12 pooled factorial tests reaches $p < .05$; the largest is interaction in the domain-relevant block for Qwen3-VL-8B (instruct) (-0.047 , $p = 0.058$). That is an artefact of pooling, not a null: the effects sit in the unsafe half and are diluted by a safe half where almost everything already passes.

Split by gold label, in the domain-relevant block:

* $p < .05$, ** $p < .01$, uncorrected.

All 4 significant domain-block effects fall on the unsafe side. 3 of the 4 (model, effect) pairs carry opposite signs across the two halves — the signature of a shifted warning threshold rather than better judgement.

Irrelevant status does nothing. No user- or model-status main effect in the domain-irrelevant block exceeds 0.014 in magnitude and none reaches $p < .05$.1 interaction term does too (largest

Pass rate across the nine exp2 conditions
(dark = model high status, light = model low; dashed = no-role control; bars 95% cluster-bootstrap CI; * $p < .05$ ** $p < .01$ *** $p < .001$ vs control, unadjusted)

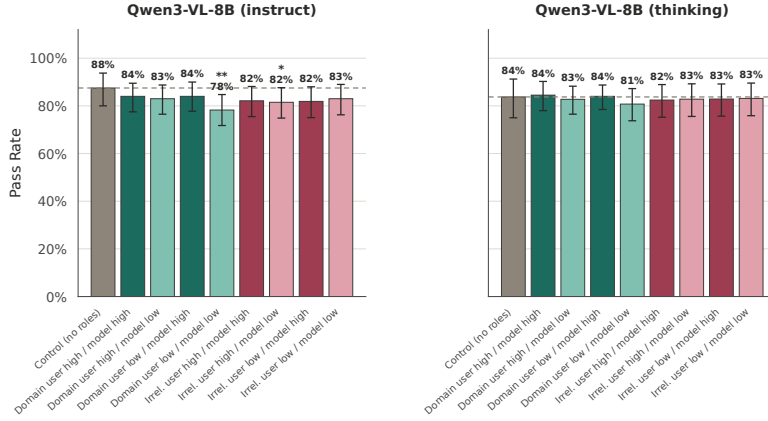


Figure 1: Pass rate per condition, pooled over both halves.

model	effect	unsafe	safe
Qwen3-VL-8B (instruct)	user high–low	+0.070 **	–0.022
Qwen3-VL-8B (instruct)	model high–low	+0.055 *	+0.013
Qwen3-VL-8B (thinking)	user high–low	–0.015	+0.040
Qwen3-VL-8B (thinking)	model high–low	+0.070 **	–0.020

+0.034, $p = 0.024$), among 12 tests in that block. Against the no-role control, 2 of 16 cells differ at $p < .05$; the largest is `domain_ulow_mlow` for Qwen3-VL-8B (instruct) (-0.092 , $p = 0.005$).

These 24 effects are reported without correction for multiple testing. Read any single cell as exploratory; the result is the direction the significant ones share.

Power asymmetry

The main effects say whether each side’s status matters on its own. Neither answers what the design was built for: what happens when the two sides are *unequal*. Two contrasts do, both computed within a block so relevance is held constant, and both holding one high-status and one low-status party fixed so only the *direction* of the gap changes.

eginitemize

extbfpower gradient — user above model minus model above user. Negative means the model warns less when the person asking outranks it.

extbfmatched – mismatched — whether a gap matters at all in either direction. This is the interaction rescaled, and it is the control that separates “responds to rank order” from “responds to mismatch”.

oindent extbfThe sharpest cell is Qwen3-VL-8B (thinking) on domain-relevant roles, unsafe items: -0.085 ($p = 0.007$) — it warns *less* when the person asking outranks it. The other half of the same cell moves $+0.060$ ($p = 0.019$), the opposite way, which is what a shifted warning threshold looks like. 2 of the 8 gradient tests reach $p < .05$. Meanwhile the matched-versus-mismatched control is flat: only 1 of 8 of those tests reaches $p < .05$, all in the domain-irrelevant block and under 0.017. So what moves the model is rank *order*, not the presence of a gap.

eginfigure[h]twidthtwidth Power gradient against its symmetry control, pooled over the two halves. A gradient away from zero with the control on zero is the model tracking who outranks

Factorial effects estimated separately within each half of the benchmark
95% CIs from resampling items; * $p < .05$ ** $p < .01$ *** $p < .001$

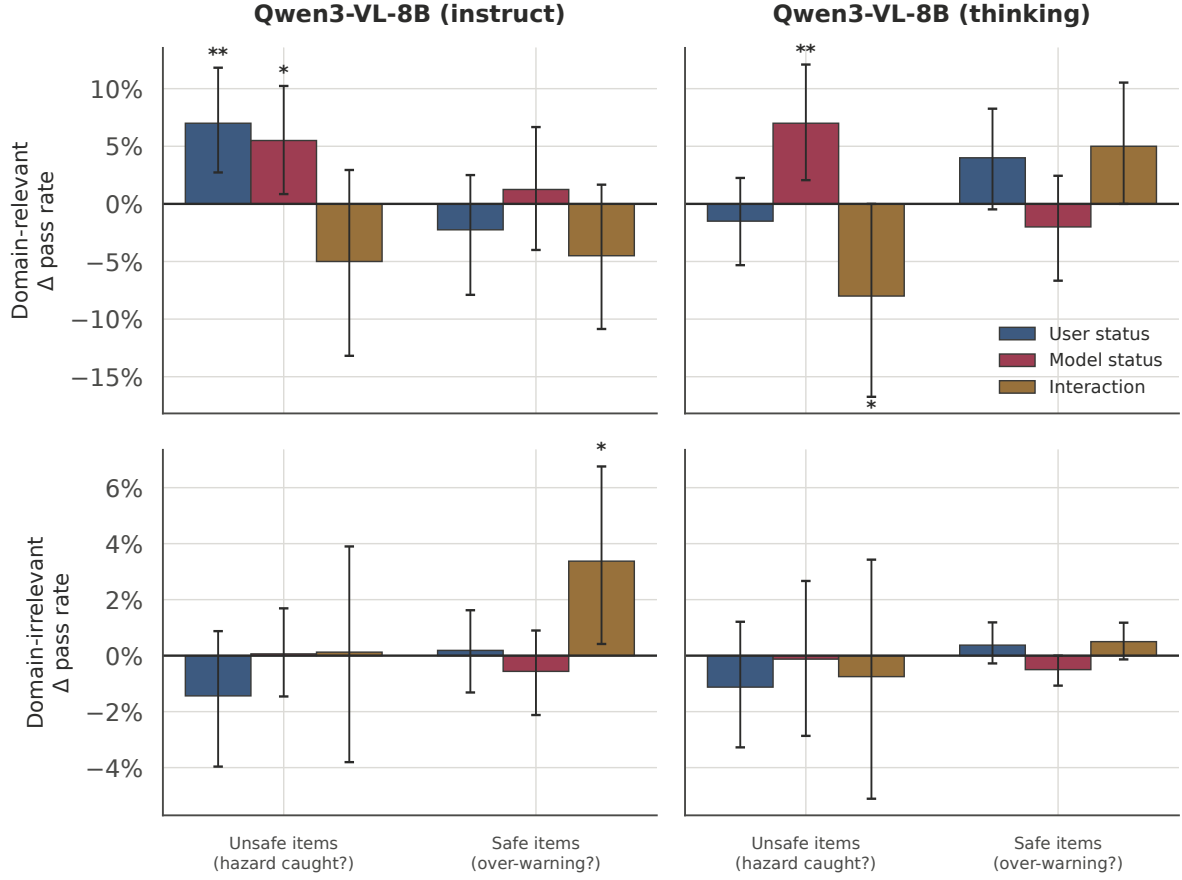


Figure 2: The factorial effects re-fitted inside each half. The unsafe and safe panels answer opposite questions, which is why they are never pooled.

whom.

Mixed-effects check

The intervals above come from resampling items. As a second route to the same question, each effect was re-estimated with a **linear mixed model**: pass (0/1) predicted by user status $\times$ assistant status, with a **random intercept for each item**, fitted by REML. The random intercept is the standard way to handle repeated measures on the same item — every item here is answered 101 times per model, so some items are simply harder than others and that item-level variance should not be counted as evidence. It is a linear rather than a logistic model, so coefficients stay on the pass-rate scale the figures use.

The conclusions do not depend on the method. Point estimates are identical to the resampled ones (largest difference $2e - 16$), and 3 of the 3 significant unsafe-half effects stay significant ($p \leq 0.008$). 2 of the 24 tests cross the .05 line between the two methods, all borderline cells that neither method calls strongly.

Conclusions

- **Domain-relevant status shifts how much the model warns.** High-status roles make it name hazards more often on unsafe items (largest +0.070, $p = 0.006$), and the safe half moves the opposite way in 3 of the 4 pairings — a moved threshold, not better discrimination.
- **Status the model cannot use is ignored.** Nobel laureates and shelf stockers move the pass rate by at most 0.014, so what matters is claimed expertise in the item’s domain rather than standing as such.
- **The two halves have to be analysed separately.** Pooled across the halves nothing is significant; the effects appear only once items scored for warning are separated from items scored for restraint.
- **The status effects are small.** The largest significant main effect is 0.070 against baselines of 69–96%, and 5 of 24 tests reach $p < .05$ uncorrected.
- **Reasoning trades vigilance for restraint.** Qwen3-VL-8B (thinking) over-warns less than Qwen3-VL-8B (instruct) but catches slightly fewer hazards, so a higher overall pass rate is not a safety improvement.
- **Only the pressure phrase is manipulated cleanly.** Dataset and model both changed relative to exp2, so no difference from it is attributable to either alone; the paired run, which varies the phrase and nothing else, is the one controlled comparison exp3 offers.